



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 06



By- Satish Sir

Recap of Previous Lecture



Topic

Types of Graph Part-02



Topics to be Covered



Topic

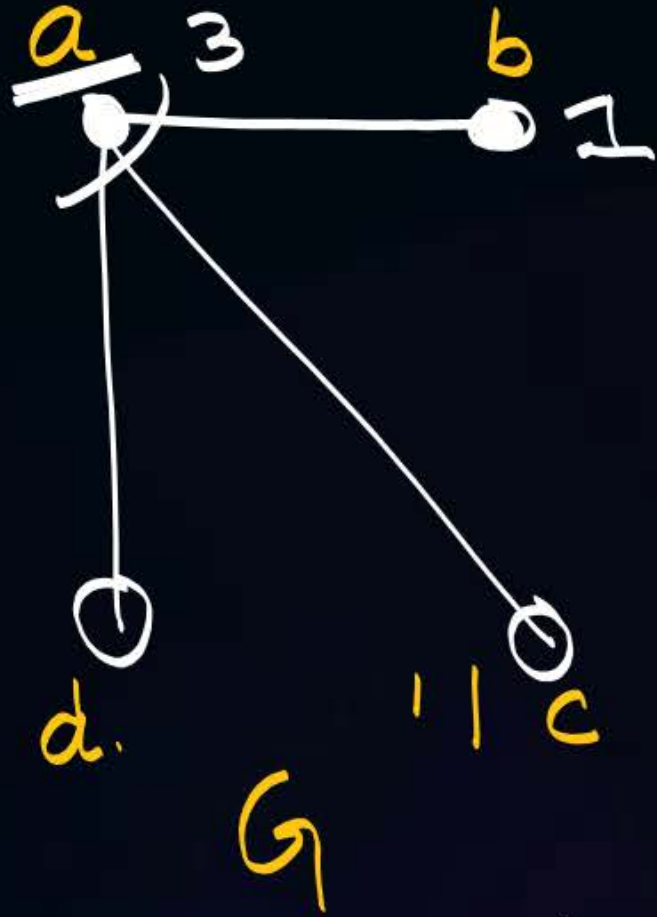
Types of Graph Part-03



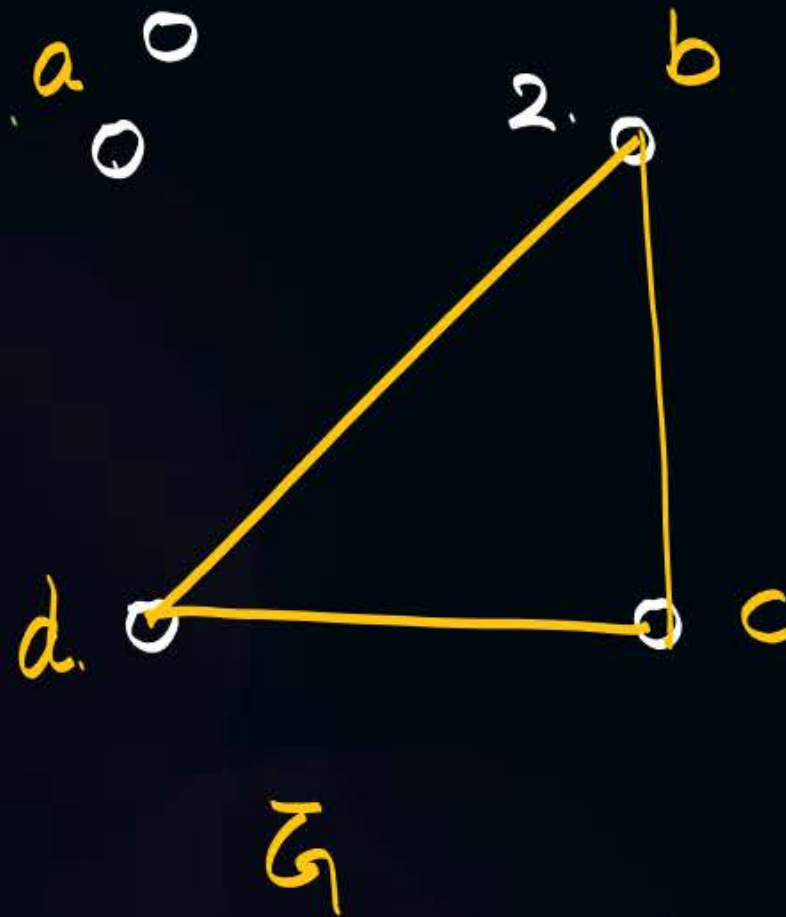


Topic : Types of Graph

Complement Graph. ($\bar{G}$)



edges $\rightarrow$ present
edges $\rightarrow$ absent



edges $\rightarrow$ absent
edges $\rightarrow$ present

$$G + \bar{G} = K_n$$

$$1) \quad e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

	a	b	c	d
K_4	3	3	3	3
$G \rightarrow$	3	1	1	1
$\bar{G} \rightarrow$	0	2	2	2

Total vertices = 4.

K_n $n-1, n-1, n-1, \dots$

$G \rightarrow d_1, d_2, d_3, d_4, \dots, d_n$ (Degree sequence)

$\overline{G}$ $n-1-d_1, n-1-d_2, n-1-d_3, n-1-d_4, \dots$

G is having degree sequence as $5, 2, 2, 2, 2, 1$.

what will be total no. of edges in $\overline{G}$?

K_6 $5, 5, 5, 5, 5, 5$

$G \rightarrow 5, 2, 2, 2, 2, 1$

Total vertices = 6

$\overline{G}$ $0, 3, 3, 3, 3, 4$

$$\sum d(v_i) = 2e$$

$$16 = 2e$$

$$e = 8$$



Topic : Types of Graph



Consider a Graph of 10 vertices & 20 edges.

What will be total edges in the complement of a Graph?

$$n = 10$$

$$e(G) = 20$$

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

↓

$$20 + e(\bar{G}) = \frac{10 \cdot 9}{2}$$

$$e(\bar{G}) = 45 - 20 \\ = 25$$

Q2: Graph is having degree sequence as

5, 2, 2, 2, 2, 1. $\{n=6\}$

What will be total edges in the complement of the above graph?

$$\sum d(v_i) = 2e$$

$$14 = 2e$$

$$e = 7$$

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

$$7 + x = \frac{6 \cdot 5}{2}$$

$$7 + x = 15$$

$$x = 8$$

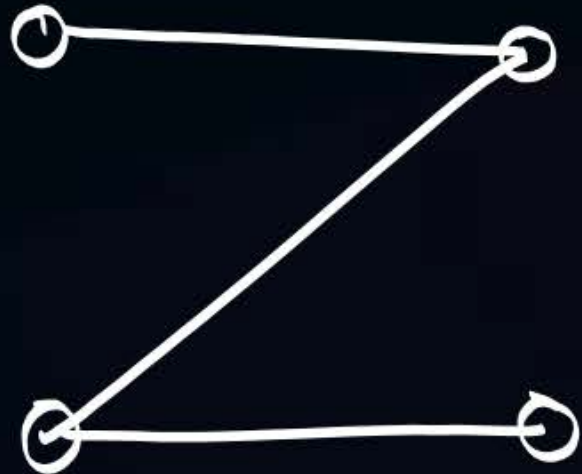


Topic : Types of Graph

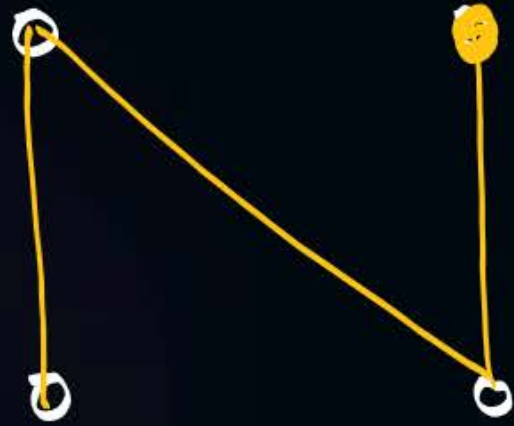


Self-complement Graph ($G \cong \bar{G}$)

if a Graph is isomorphic to its own complement, then it is self-complement Graph.



G



$\bar{G}$

$$1) e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

$$\downarrow \quad \downarrow \\ e + e = \frac{n(n-1)}{2}$$

$$2e = \frac{n(n-1)}{2}$$

$$e = \frac{n(n-1)}{4}$$



Topic : Types of Graph



$$e = \frac{n(n-1)}{2}$$

$$\frac{n-0}{2} \text{ or } \frac{n-1}{2}$$

$$n=4 \quad e = \frac{n(n-1)}{2} = \frac{4 \cdot 3}{2} = 3 \quad \checkmark$$

$$n=5 \quad e = \frac{5 \cdot 4}{2} = e = 5 \quad \checkmark$$

$$n=6 \quad e = \frac{6 \cdot 5}{2} = \frac{15}{2} \quad \times$$

$$n=7 \quad e = \frac{7 \cdot 6}{2}$$

$$\underline{n \equiv 0 \pmod{2} \text{ or } n \equiv 1 \pmod{2}}$$

$$n \equiv 0 \text{ or } 1 \pmod{2}$$

1. $a \equiv b \pmod{n}$ 3. a, b are having same remainder w.r.t n . $(1 = 2 = 3)$

2. $\frac{a-b}{n} \in \mathbb{Z}$

$1 \equiv 5 \pmod{4}$

$\frac{1-5}{4} \in \mathbb{Z}$

$n=4$

$0 \equiv 4 \pmod{4}$ (remainder '0')



Topic : Types of Graph

what will be value of n in C_n , where it will become self-complement.

Set of cycle Graph.

C_3	X
C_4	X
C_5	
C_6	X
C_7	X
C_8	X
C_9	X

$$e = \frac{n(n-1)}{4}$$
$$= \frac{10 \cdot 9}{4}$$
$$= 10$$

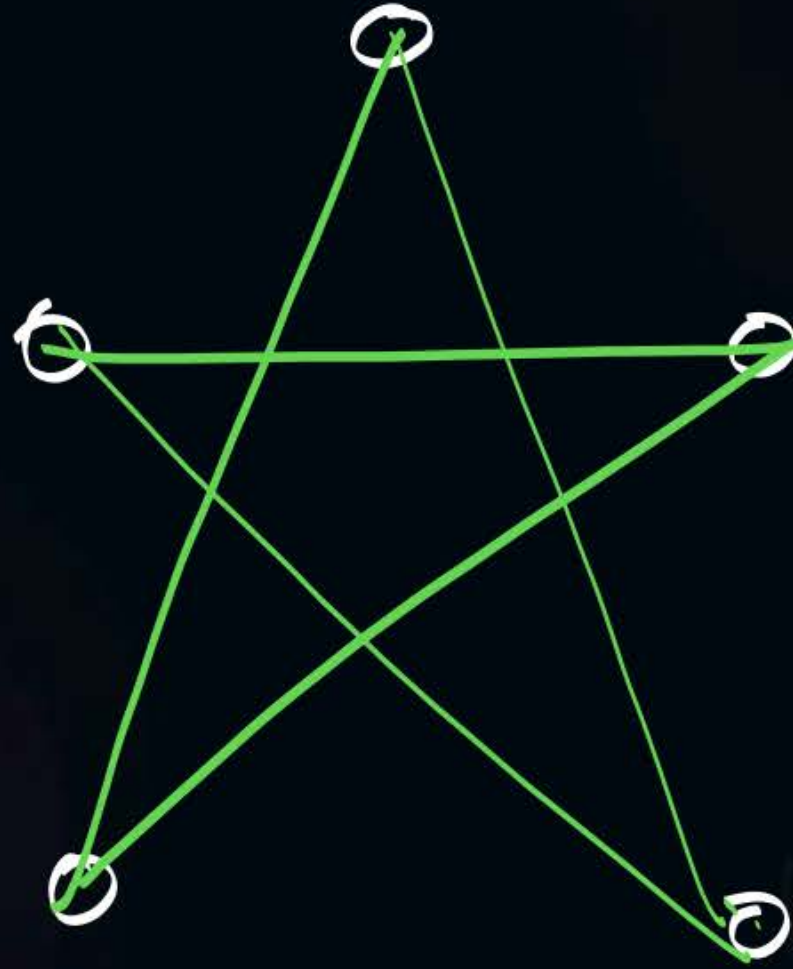
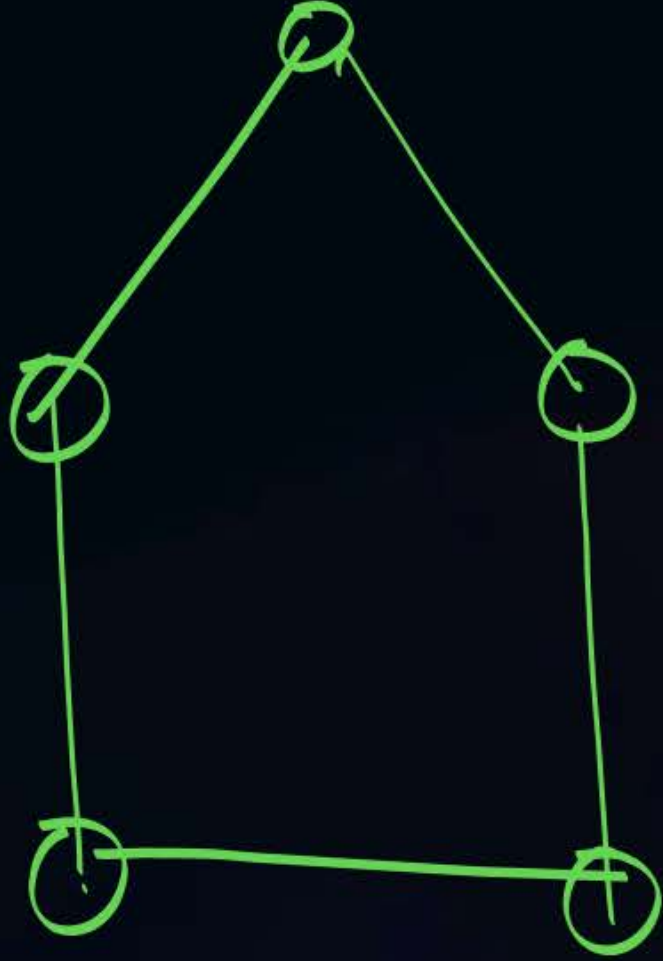
Assume C_8 is self complement

that means in C_8

$$n = e = 8$$

self complement Ans: 5.

$$e = \frac{n(n-1)}{4}$$
$$= \frac{8 \cdot 7}{4} = 14$$



G_5

$$\begin{cases} n = e = 5. \\ 2, 2, 2, 2, 2. \end{cases}$$



Topic : Types of Graph

Isomorphic : (Same Graph, but diff representation)

Two Graphs G_1, G_2 are isomorphic to each other if they have same incident property.

(meeting property)

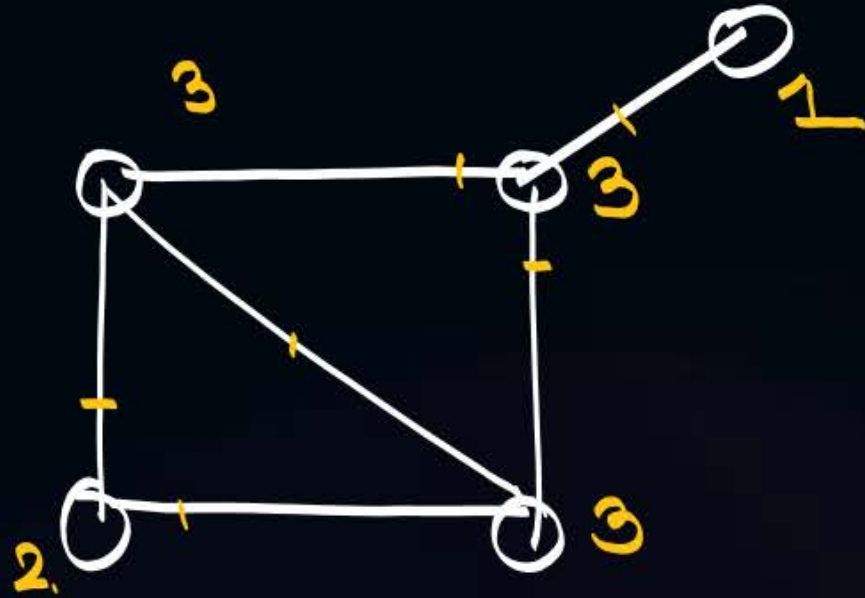
→ no. of vertices

→ no. of edges

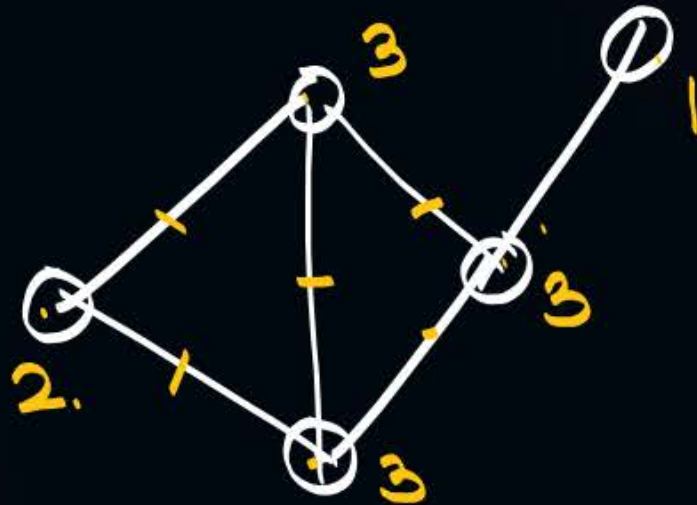
→ Degree sequence



Topic : Types of Graph



$$\begin{cases} n = 5 \\ e = 6 \\ 3 \ 3 \ 3 \ 2 \ 1 \end{cases}$$



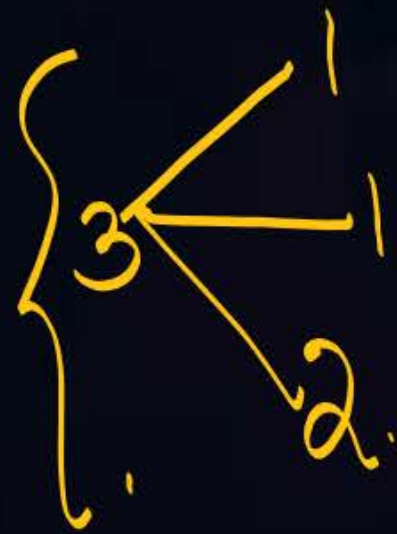
$$\begin{cases} n = 5 \\ e = 6 \\ 2 \ 3 \ 3 \ 3 \ 1 \end{cases}$$



Topic : Types of Graph



$$\begin{aligned}n &= 6 \\e &= 5 \\3 \ 2 \ 2 \ 1 \ 1 \ 1\end{aligned}$$



$$\begin{aligned}n &= 6 \\e &= 5 \\3 \ 2 \ 2 \ 1 \ 1 \ 1\end{aligned}$$

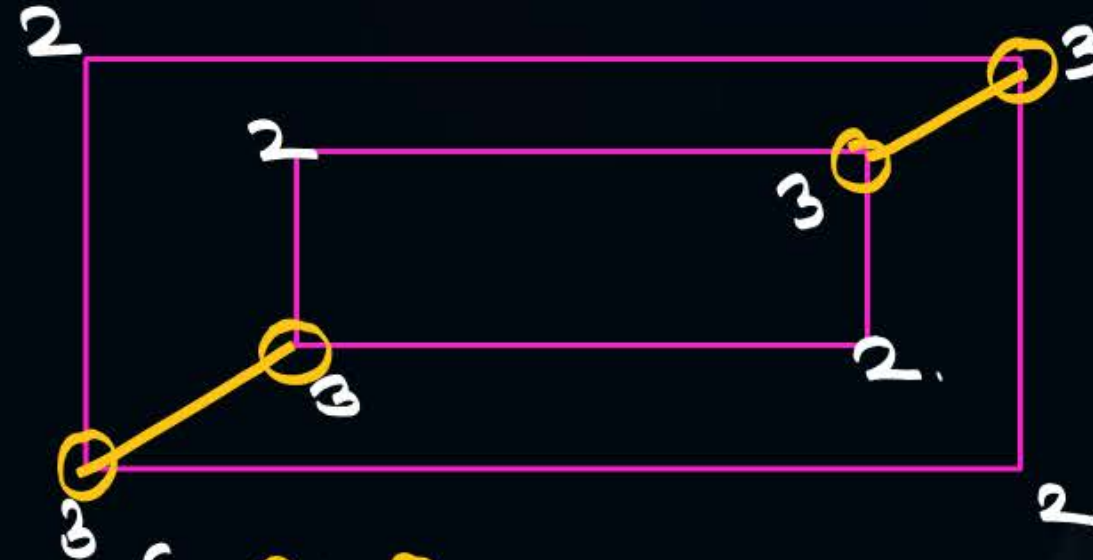




Topic : Types of Graph



$$\left\{ \begin{array}{l} n=8 \\ e=10 \\ 33332222 \end{array} \right.$$

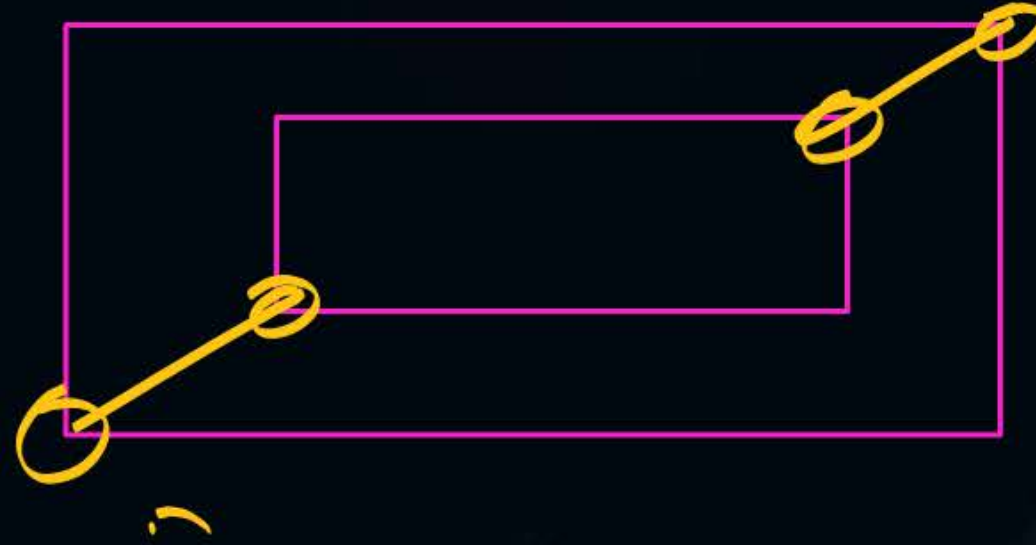


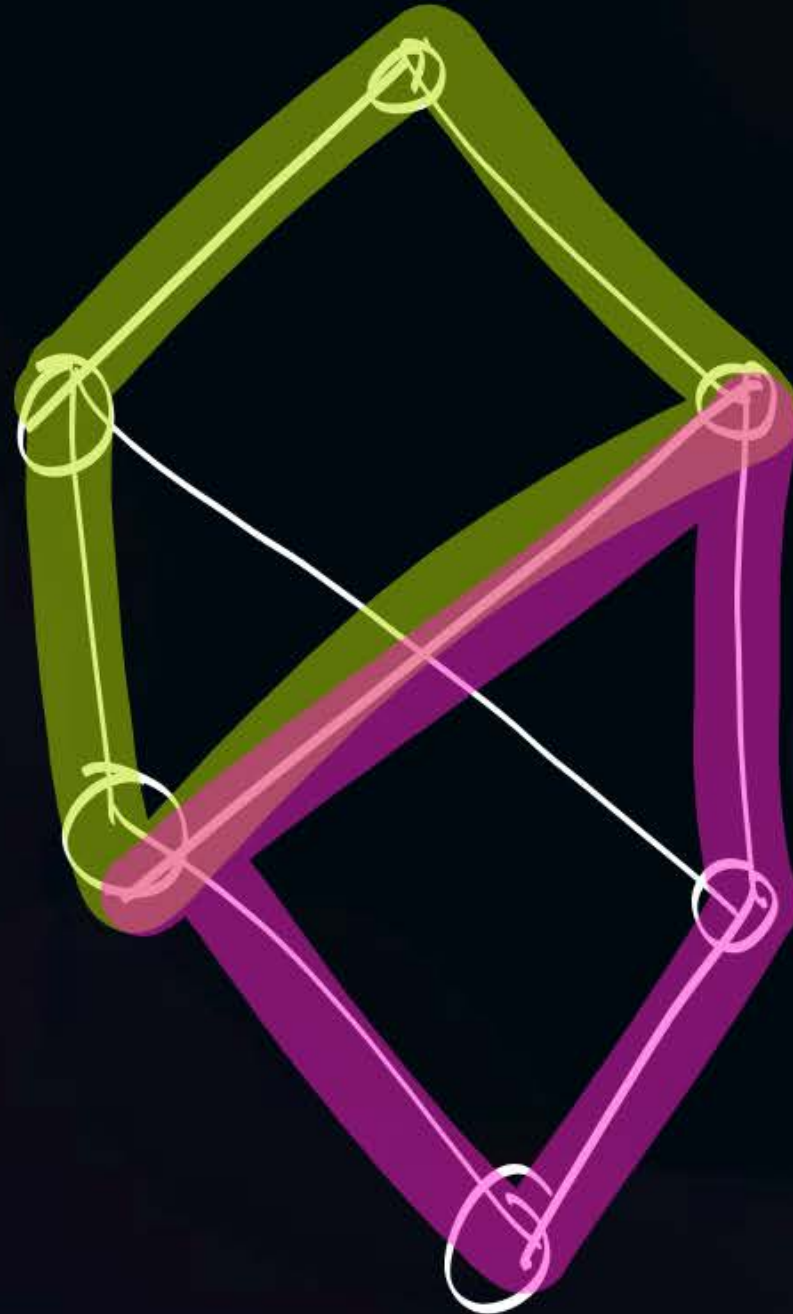
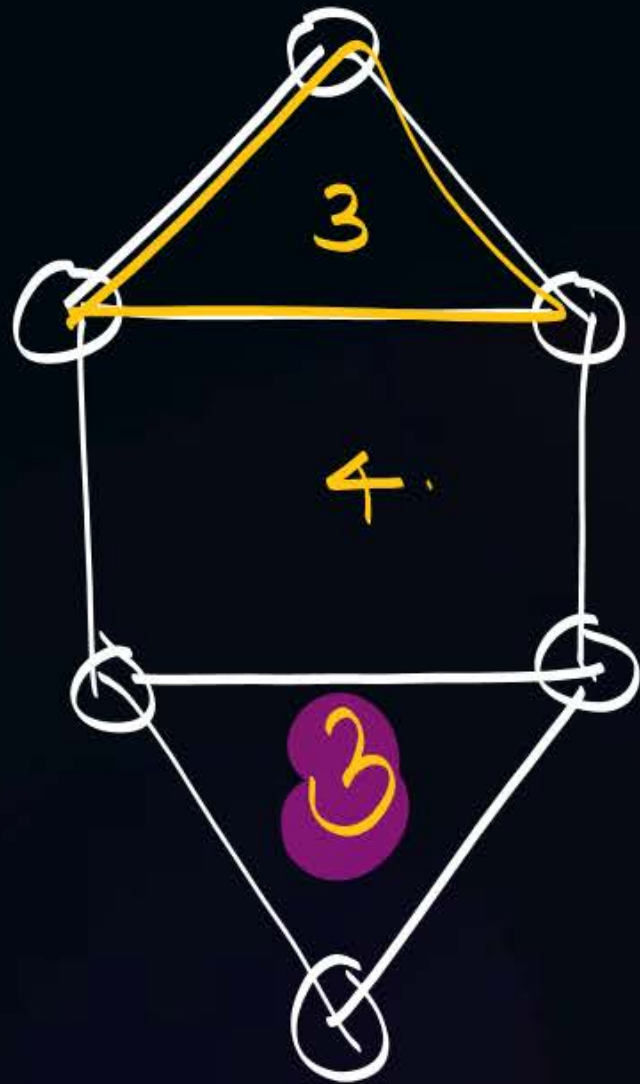
$$\left\{ \begin{array}{l} n=8 \\ e=10 \\ 33332222 \end{array} \right.$$

Cycle
length



Topic : Types of Graph





Rosen.
 $\downarrow$
isomorphic.
 $\downarrow$
 Question.



Topic : Types of Graph

Graph:

$$G = (V, E)$$

* each edge must be an association in order pair of vertices.

eg:



$$G = (V, E)$$

$$V = \{a, b\}$$

$$E = \{e_1\}$$

$$* e_1 \rightarrow (a, b)$$

Defn:

$$G = (V, E, \delta) \quad \text{incident function}$$

$$* \delta: E \rightarrow V \times V$$



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Two Graphs G_1, G_2 are isomorphic to each other if they have 1:1 correspondance function.

$$f: G_1 \rightarrow G_2$$

$$G_1 = (V_1, E_1, \delta_1)$$

$$G_2 = (V_2, E_2, \delta_2)$$

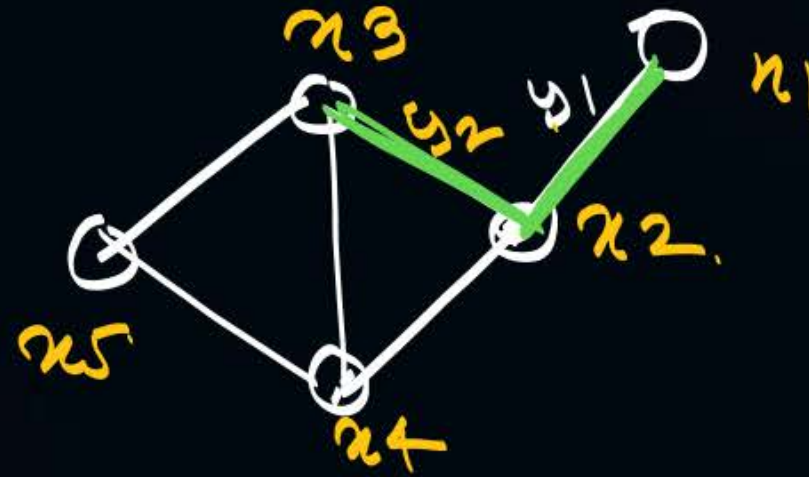
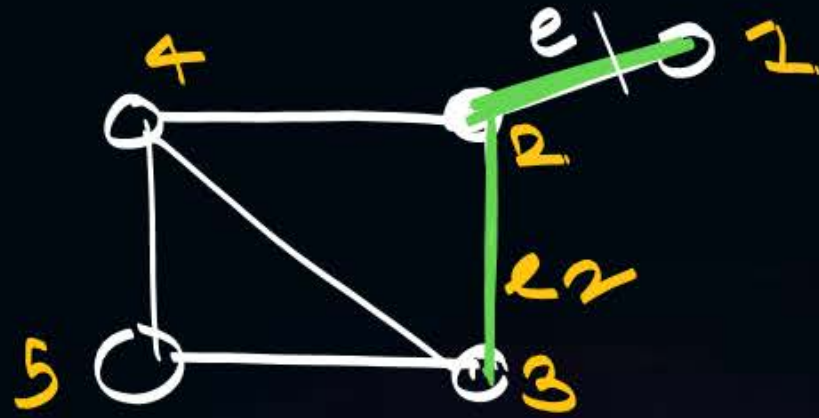
$$f: V_1 \rightarrow V_2$$

$$f: E_1 \rightarrow E_2$$

$$f: \delta_1 \rightarrow \delta_2$$



Topic : Types of Graph



$$f: \delta_1 \rightarrow \delta_2$$

$$e_1 \rightarrow (1, 2) \quad y_1 \rightarrow (x_1, x_2)$$

$$f: v_1 \rightarrow v_2$$

$$1 \rightarrow x_1$$

$$2 \rightarrow x_2$$

$$3 \rightarrow x_3$$

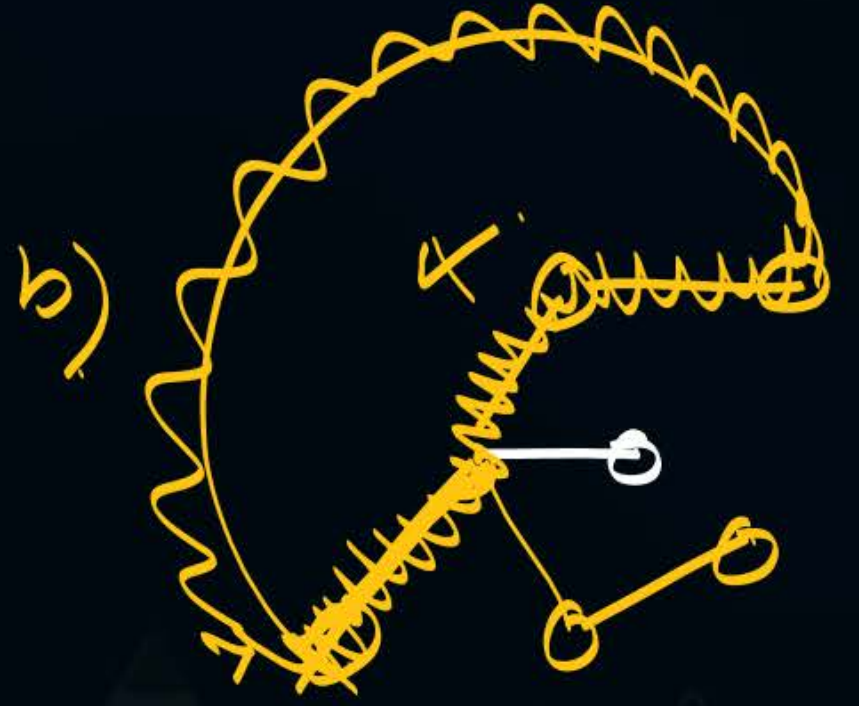
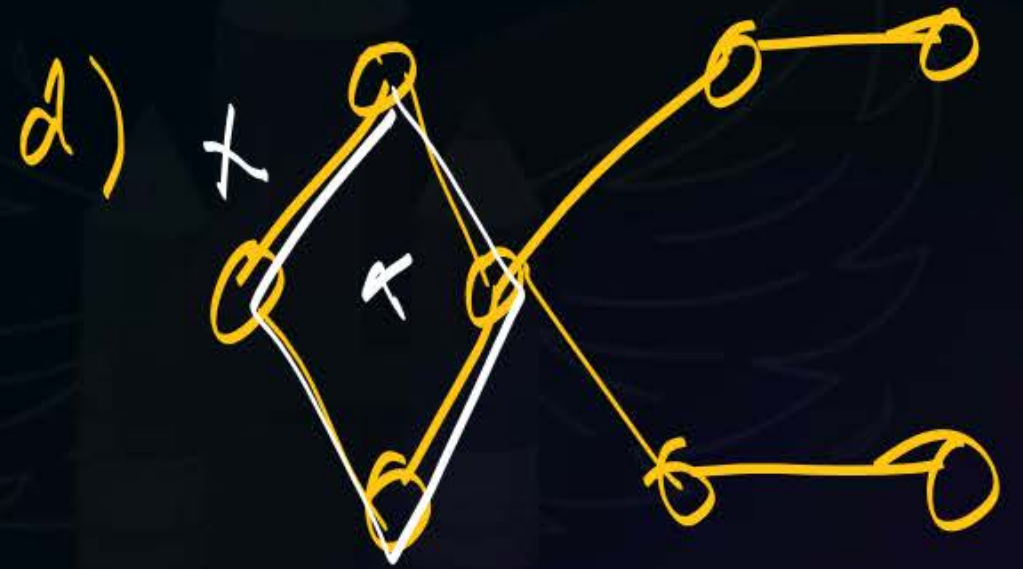
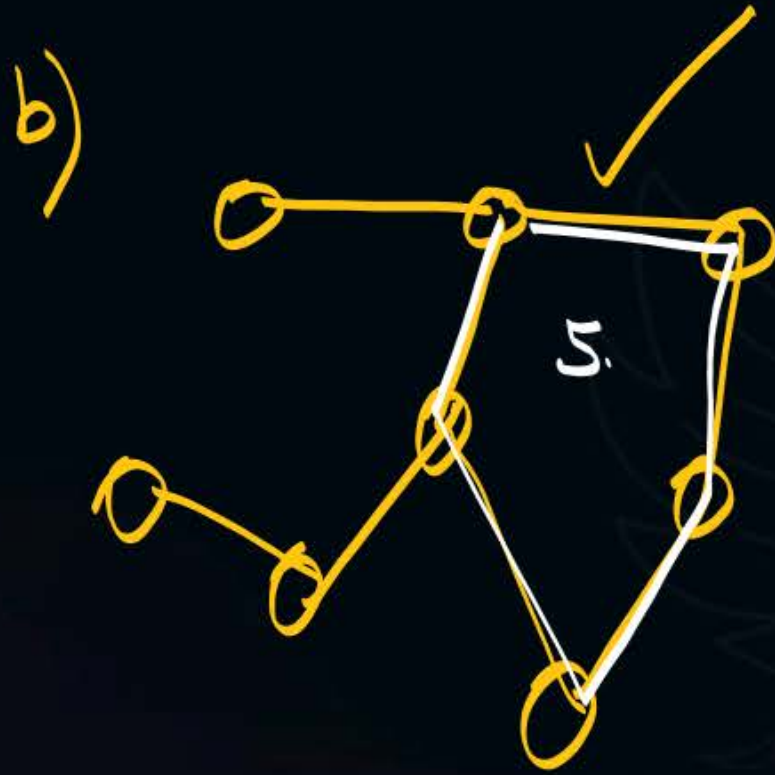
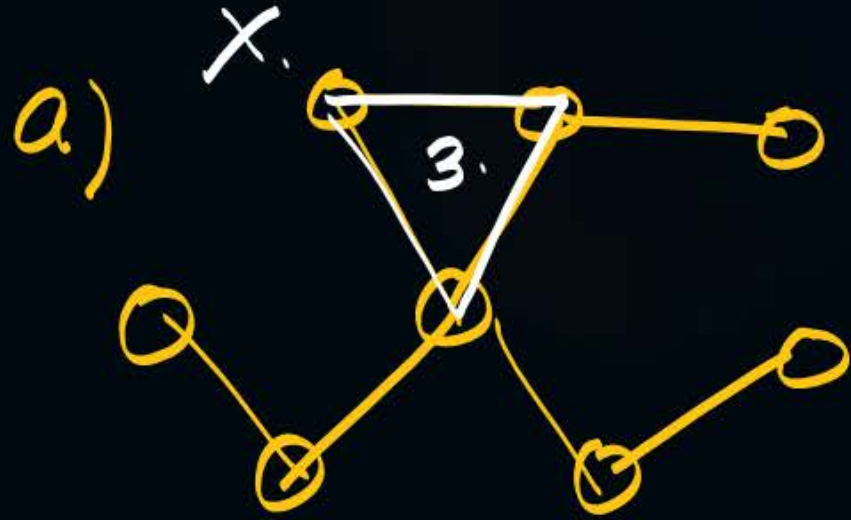
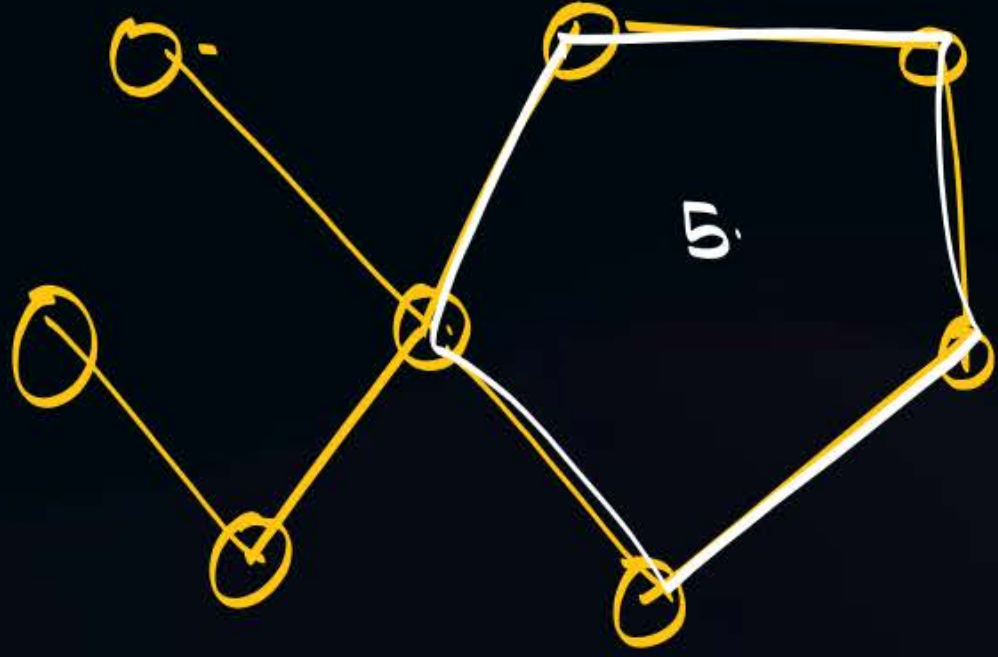
$$4 \rightarrow x_4$$

$$5 \rightarrow x_5$$

$$f: E_1 \rightarrow E_2$$

$$e_1 \rightarrow y_1$$

$$e_2 \rightarrow y_2$$

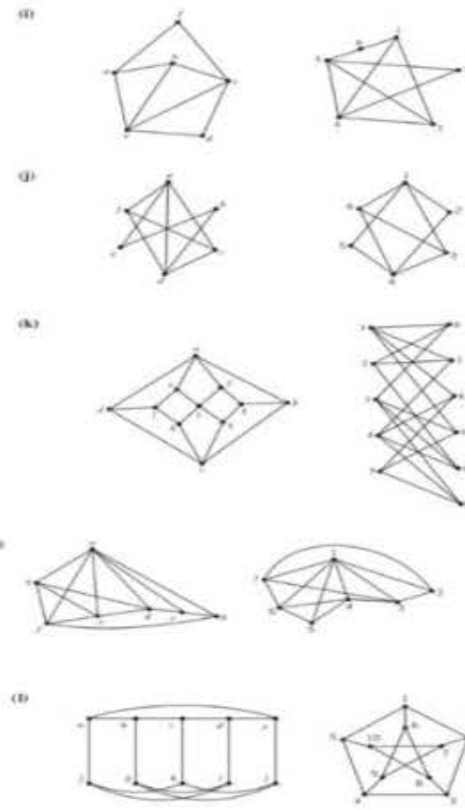


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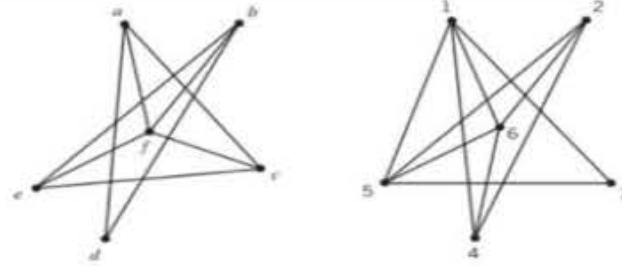
LTE

GRAPH PART1.pdf

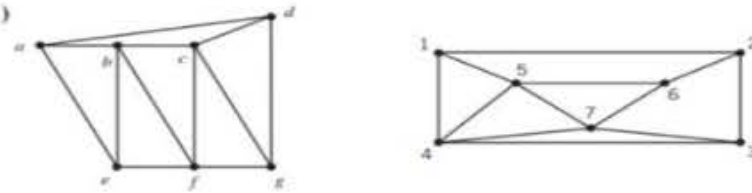
Done



(e)



(f)



(g)





2 mins Summary

Topic

One

Types of Graph Part-03

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU